

RETHINKING KOREAN HOW-QUESTIONS: A CROSS-LINGUISTIC STUDY*

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ABSTRACT

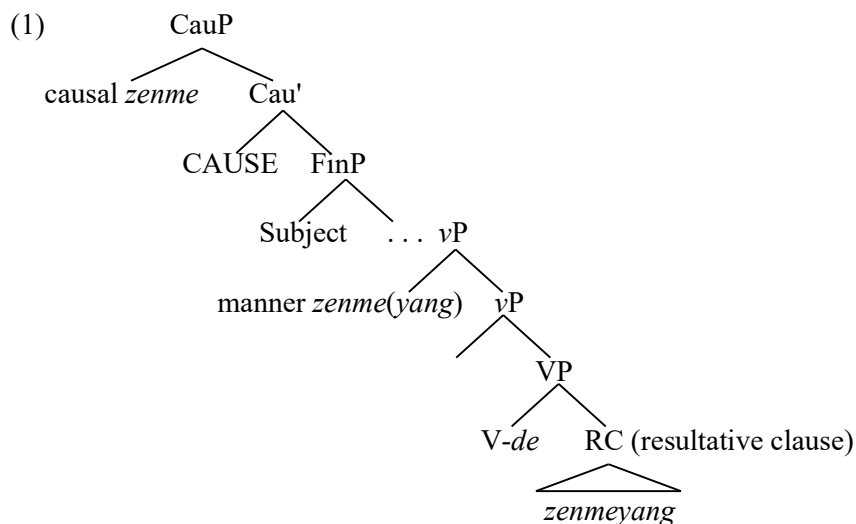
Ettehkey ‘how’ in Korean can be interpreted as either causal, manner, or resultative *how* according to its “height of interpretation” (Hacquard 2006). Interestingly, it is immune from strong island effects when interpreted as manner or resultative, whereas it is island-sensitive when interpreted as causal. This observation presents a challenge to the Noun versus Adverb Generalization of *wh*-in-situ construals (Tsai 1999a; Fujii et al. 2014), since *ettehkey* does not seem to contain a nominal head. In this paper, it is argued that the difficulty is only apparent, and the unselective binding analysis can be extended to Korean *how*-questions with manner and resultative *ettehkey* being decomposed into a couple of functional elements such as an indeterminate noun (Kuroda 1965), whereas Korean causal *how* has evolved into a genuine adverb in the left periphery. We also draw cross-linguistic evidence from Chinese and Japanese to further support this novel approach in combining the insights from the Syntactic Cartography and the Distributed Morphology.

Keywords: comparative syntax, unselective binding, grammaticalization, Syntactic Cartography, Distributed Morphology, *wh*-in-situ

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1. INTRODUCTION

Chinese has various types of *how*-expressions. Among them, causal, manner, and resultative *how* exhibit clear distributional and morphological differences as illustrated in (1). They are respectively situated at Spec,Cau(se)P, Spec,vP, and the complement position of the VP according to many previous studies (Stepanov & Tsai 2008; Tsai 1999b, 2008, 2023, 2024).



Korean also features these types of *how* that are respectively distributed in three different positions (cf. O. Kim 2021; 2022 and O. Kim and S. Park 2021):

- (2) a. ne-ka **ettehkey** pantusi kheyikhu-lul mantul-eyaha-ni?
 you-NOM **how** necessarily cake-ACC make-must-Q
 ‘What is the cause x such that, due to x, you necessarily have to make a cake?’
 *‘What is manner x such that, in x, you necessarily have to make a cake?’
 *‘What is the result x such that, into x, you necessarily have to make a cake?’

- b. ne-ka pantusi **ettehkey** kheyikhu-lul mantul-eyaha-ni?
 you-NOM necessarily **how** cake-ACC make-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to make a cake?’
 ‘What is manner x such that, in x, you necessarily have to make a cake?’
 *‘What is the result x such that, into x, you necessarily have to make a cake?’
- c. ne-ka pantusi kheyikhu-lul **ettehkey** hay-yaha-ni?¹
 you-NOM necessarily cake-ACC **how** do-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to do something with the cake?’
 *‘What is manner x such that, in x, you necessarily have to do something with the cake?’
 ‘What is the result x such that, into x, you necessarily have to make a cake?’

Unlike Chinese (1), however, the three *how*-construals in Korean appear to have a single morphological form due to their undistinguishable phonetic form, i.e., *ettehkey*. Nevertheless, we can still analyze their morpho-syntactic make-ups into some structures different from each other, based on various kinds of diagnostics and syntactic properties they exhibit, such as island sensitivity, scope relations, availability of alternatives, and so on.

In Section 2, we introduce O. Kim and S. Park’s (2021) data and arguments about the Korean *how*-construals and challenge them with some possible counter-examples to their assumptions, causal *eccayse* and non-causal *ettehakey*. Section 3 introduces the unselective binding analysis (Tsai, 1999a; 2008) as a syntactic diagnostic and discusses the constituencies of causal *eccayse* and *ettehkey* as genuine *wh*-adverbs

¹ Thankfully, one of the reviewers has kindly inquired whether resultative reading of (2b) is still possible. However, (2c) is a typical example of resultative *how*, and *ettehkey* in this case cannot appear before the internal argument as exemplified below:

- (i) *ne-ka pantusi **ettehkey** kheyikhu-lul hay-yaha-ni?
 you-NOM necessarily **how** cake-ACC do-must-Q
 ‘What is the result x such that, into x, you necessarily have to make a cake?’

We take this as the strongest evidence for the relatively low position of resultative *ettehkey* (see also Nam 1988 and Nam and Cho 1993 for further examples like this). Therefore, we still judge resultative reading of (2b) as ungrammatical.

positioned at the specifier position of CauP, an outer light verb phrase (Tsai 2014; 2015). In Section 4, we etymologically analyze and account for the two non-causal *ettehkey* (2b, c) from the perspective of Distributed Morphology (Halle and Marantz 1993; Marantz 1994; and Embick and Noyer 2007). Section 5 summarizes the arguments of this paper.

2. SETTING THE PROBLEM

2.1 O. Kim and S. Park's (2021) arguments

According to O. Kim and S. Park (2021:808), causal *ettehkey* (3a) and manner *ettehkey* (3b) exhibit an asymmetry in island sensitivity. We agree with their grammaticality judgements, but, in fact, resultative *ettehkey* is also island-insensitive (3c).

- (3) Mimi-nun [NP_{CP} Kim-i **ettehkey** yoliha-n] umsik]-ul
 Mimi-TOP Kim-NOM **how** cook-MOD food-ACC
 mekess-ni?
 ate-Q
 a. *‘What was the cause x such that Mimi ate the food that, due to x, Kim cooked?’
 b. ‘What was manner x such that Mimi ate the food that, in x, Kim cooked?’
 c. ‘What was the result x such that Mimi ate the food that, into x, Kim cooked?’

Based on the Intervention Effect asymmetry between causal and non-causal *ettehkey*, O. Kim and S. Park further argue that causal *ettehkey* is positioned in the CP-layer. For example, in Korean, *wh*-arguments such as *mwe* ‘what’ cannot be associated with the CP-layer across a scope-bearing element (SBE), such as *amwuto* ‘anyone’ (Beck & Kim 1997; Beck, 2006). Consider the following example:

- (4) *amwuto **mwe-l** ilk-ciahn-ass-ni?
 anyone **what**-ACC read-NEG-PST-Q
 ‘What did no one read?’

In contrast, the *wh*-expression *way* ‘why’ is immune to Intervention Effect caused by the SBE as illustrated in the following example:

- (5) {**way**} amwuto {**way**} nonmwun-ul ilk-ciahn-ass-ni?
 why anyone **why** paper-ACC read-NEG-PST-Q
 ‘Why did no one read the paper?’

Ko (2005, 2006) argues that the immunity of *way* ‘why’ to the Intervention Effect is due to its base position. Specifically, according to Ko, *way* ‘why’ is base-generated at Spec,CP, higher than the SBE, situated at Spec,IP, whereas *mwe* ‘what’ in (4) is base-generated under the scope of the SBE. Consequently, *way* ‘why’ does not establish any dependency across the SBE and hence becomes immune to the Intervention Effect. This claim is further supported by the observation that the SBE must be scoped over by *way* ‘why’ in semantics, even when it is scrambled over *way* ‘why’ (5).

According to O. Kim and S. Park, causal *ettehkey* also exhibits a similar pattern, whereas the two non-causal *ettehkey* do not. (cf. Choi 2007). Consider the following examples:

- (6) a. **ettehkey** amwuto kheyikhu-lul mantul-ciahn-ass-ni?
 how anyone cake-ACC make-NEG-PST-Q
 ‘What was the cause x such that, due to x, nobody made any cake?’
 *‘What was manner x such that, in x, nobody made any cake?’
 *‘What was the result x such that, into x, nobody made any cake?’
 b. amwuto **ettehkey** kheyikhu-lul mantul-ciahn-ass-ni?
 anyone **how** cake-ACC make-NEG-PST-Q
 ‘What was the cause x such that, due to x, nobody made any cake?’
 *‘What was manner x such that, in x, nobody made any cake?’²
 *‘What was the result x such that, into x, nobody made any cake?’

² Manner and resultative interpretations (6b, c) are possible when the relevant SBEs are destressed. However, SBEs can be destressed only when the relevant proposition, i.e., ‘nobody made any cake’, is already given in the Common Ground.

- c.* *amwuto* kheyikhu-lul **ettehkey** ha-ciahn-ass-ni?
anyone cake-ACC **how** do-NEG-PST-Q
 *‘What was the cause x such that, due to x, nobody did anything with the cake?’
 *‘What was manner x such that, in x, nobody did anything with the cake?’
 *‘What was the result x such that, into x, nobody made the cake into?’²

(6b) and (6c) demonstrate that the non-causal interpretations are consistently blocked under the scope of the SBE. In contrast, causal *ettehkey* remains unaffected, indicating its position outside the scope of the SBE. Based on these observations, O. Kim and S. Park claim that causal *ettehkey* is positioned in the CP-layer, akin to *way* ‘why’, whereas manner *ettehkey* is positioned below the CP-layer.

We can further support O. Kim and S. Park’s argument with our own data. The two non-causal *ettehkey* are sensitive to negative islands as well, whereas causal *ettehkey* is not. Consider the following examples:

- (7) a. ne-ka **ettehkey** pantusi kheyikhu-lul
 you-NOM **how** necessarily cake-ACC
 mantul-cianh-ulswukaiss-ni?
 make-NEG-can-Q
 ‘What is the cause x such that, due to x, necessarily you may not make a cake?’
 *‘What is manner x such that, in x, necessarily you may not make a cake?’
 *‘What is the result x such that, into x, necessarily you may not make a cake?’
- b.* ne-ka pantusi **ettehkey** kheyikhu-lul
 you-NOM necessarily **how** cake-ACC
 mantul-cianh-ulswukaiss-ni?
 make-NEG-can-Q
 *‘What is the cause x such that, due to x, necessarily you may not make a cake?’

*‘What is manner x such that, in x, necessarily you may not make a cake?’

*‘What is the result x such that, into x, necessarily you may not make a cake?’

- c. *ne-ka pantusi kheyikhu-lul **ettehkey**
 you-NOM necessarily cake-ACC **how**
 ha-cianh-ulswukiss-ni?
 do-NEG-can-Q

*‘What is the cause x such that, due to x, necessarily you may not do anything with the cake?’

*‘What is manner x such that, in x, necessarily you may not do anything with the cake?’

*‘What is the result x such that, into x, necessarily you may not make a cake?’

Given that Korean is strictly head-final and the Neg head *-cianh* is analyzed to be positioned immediately above the VP-layer (cf. Jung 1992), we can attribute the negative island sensitivity of the non-causal *ettehkey* (7b-c) to the Relativized Minimality effect between the quantificational features of the negation *-cianh* and the non-causal *ettehkey* (Rizzi 2004). On the other hand, causal *ettehkey* does not trigger any Relativized Minimality effect. Furthermore, the semantic interpretation of causal *ettehkey* (7a) implies that causal *ettehkey* must scope over the epistemic modal verb *-ulswukaiss* ‘may’. These results are not surprising if we regard causal *ettehkey* as a CP-layer adverb as O. Kim and S. Park do.

The scope relations between the three *ettehkey* and subject quantifier also support their argument. Causal *ettehkey* always semantically scopes over subject quantifier whereas the non-causal *ettehkey* are scoped over by subject quantifier. Consider the following examples:

- (8) a. **ettehkey** motwutul kheyikhu-lul mantul-ess-ni?
how everyone cake-ACC make-PST-CONN
 ‘What was the cause x such that everyone made the cake(s) due to x?’ (how > $\forall$)
 *‘What was manner x such that everyone made the cake(s) in x?’
 ($\forall$ > how)

- *‘What was the result x such that everyone made the cake(s) into x?’ ($\forall > \text{how}$)
- b. motwutul **ettehkey** kheyikhu-lul mantul-ess-ni?
 everyone **how** cake-ACC make-PST-CONN
 *‘What was the cause x such that everyone made the cake(s) due to x?’ ($\text{how} > \forall$)
 ‘What was manner x such that everyone made the cake(s) in x?’ ($\forall > \text{how}$)
 *‘What was the result x such that everyone made the cake(s) into x?’ ($\forall > \text{how}$)
- c. motwutul kheyikhu-lul **ettehkey** hay-ss-ni?
 everyone cake-ACC **how** do-PST-CONN
 *‘What was the cause x such that, due to x, everyone did something with the cake?’ ($\text{how} > \forall$)
 *‘What was manner x such that, in x, everyone did something with the cake?’ ($\forall > \text{how}$)
 ‘What was the result x such that, into x, everyone made the cake?’ ($\forall > \text{how}$)

Therefore, (6-8) uniformly indicate that causal *ettehkey* is positioned in the CP-layer, whereas two non-causal *ettehkey* are positioned in the VP-layer, aligning with O. Kim and S. Park’s (2021) argument.

Under the assumption that causal and manner *ettehkey* are morpho-syntactically undistinguishable, O. Kim and S. Park argue that their base positions determine their island (in)sensitivity. Specifically, they claim that the island sensitivity of causal *ettehkey* (3a) arises from its base position in the CP-layer, whereas manner *ettehkey* is island-insensitive due to its base position below the CP-layer. Building on this idea, O. Kim and S. Park contend that not only Korean *ettehkey*, but also Chinese causal *zenme* ‘how’ and reason *weishenme* ‘why’ and Japanese causal *doosite* ‘how’ and reason *naze* ‘why’ exhibit island-sensitivity due to their base positions in the CP-layer, directly challenging the Noun versus Adverb Generalization (9) for *wh*-in-situ languages, which dates back to Huang (1982a; 1982b).

- (9) Noun versus Adverb Generalization (NAG)
 An in-situ *wh*-phrase is island-insensitive iff it is (or contains) a *wh*-nominal. (Fujii et al. 2014)

2.2 Some counter-examples

However, we believe that we have to reconsider O. Kim and S. Park's conclusion. Specifically, the data (6-8) are insufficient to conclude that it is the base positions, not the morpho-syntactic make-ups, that determine the island (in)sensitivity of *ettehkey*. If the base positions were the determinant of island sensitivity, and if causal and non-causal *ettehkey* were encoded by a single *wh*-adverb as O. Kim and S. Park argue, then other types of *how*-expressions in Korean would also be able to freely appear in the relevant positions with the relevant three interpretations, exhibiting the same asymmetry in island (in)sensitivity. However, this is not the case as will be described below.

Korean has various kinds of *how*-expressions, such as *eccayse* and *ettehakey*. Interestingly, unlike *ettehkey*, which is ambiguous between the three interpretations (2), *eccayse* is exclusively interpreted as causal *how*, whereas *ettehakey* is exclusively interpreted as either of the two non-causal *how*. Consider the following examples:

- (10) a. ne-ka **eccayse** pantusi kheyikhu-lulmantul-eyaha-ni?
 you-NOM **how** necessarily cake-ACC make-must-Q
 'What is the cause x such that, due to x, you necessarily have to make a cake?'
 *'What is manner x such that, in x, you necessarily have to make a cake?'
 *'What is the result x such that, into x, you necessarily have to make a cake?'
 b. *ne-ka pantusi **eccayse** kheyikhu-lulmantul-eyaha-ni?
 you-NOM necessarily **how** cake-ACC make-must-Q
 *'What is the cause x such that, due to x, you necessarily have to make a cake?'
 *'What is manner x such that, in x, you necessarily have to make a cake?'
 *'What is the result x such that, into x, you necessarily have to make a cake?'

- c. *ne-ka pantusi kheyikhu-lul **eccayse** hay-yaha-ni?
 you-NOM necessarily cake-ACC **how** do-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to do something with the cake?’
 *‘What is manner x such that, in x, you necessarily have to do something with the cake?’
 *‘What is the result x such that, into x, you necessarily have to make the cake?’

- (11) a. *ne-ka **ettehakey** pantusi kheyikhu-lul
 you-NOM **how** necessarily cake-ACC
 mantul-eyaha-ni?
 make-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to make a cake?’
 *‘What is manner x such that, in x, you necessarily have to make a cake?’
 *‘What is the result x such that, into x, you necessarily have to make a cake?’
- b. ne-ka pantusi **ettehakey** kheyikhu-lul
 you-NOM necessarily **how** cake-ACC
 mantul-eyaha-ni?
 make-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to make a cake?’
 *‘What is manner x such that, in x, you necessarily have to make a cake?’
 *‘What is the result x such that, into x, you necessarily have to make a cake?’
- c. ne-ka pantusi kheyikhu-lul **ettehakey** hay-yaha-ni?
 you-NOM necessarily cake-ACC **how** do-must-Q
 *‘What is the cause x such that, due to x, you necessarily have to do something with the cake?’
 *‘What is manner x such that, in x, you necessarily have to do something with the cake?’
 *‘What is the result x such that, into x, you necessarily have to make the cake?’

As (10) exemplifies, *eccayse* cannot be interpreted as either of the two non-causal *how* regardless of how deeply it is embedded in the VP-layer. On the contrary, *ettehakey* exhibits exactly the opposite pattern. It is exclusively interpreted as either of the two non-causal *how* within the VP-domain (cf. §2.1) and cannot be interpreted as causal *how*, even if positioned in the CP-layer as shown in (11a).

In addition, *eccayse* is island-sensitive (12a) and scopes over the subject SBE (13a), negative islands (14a) and the subject quantifier (15a), whereas *ettehakey* exhibits the opposite patterns. Consider the following examples (12-15):

- (12) a. *Mimi-nun [_{NP}[_{CP} Kim-i **eccayse** yoliha-n] umsik]-ul
 Mimi-TOP Kim-NOM **how** cook-MOD food-ACC
 mekess-ni?
 ate-Q
 *‘What was the cause x such that Mimi ate the food that, due to x, Kim cooked?’
- b. Mimi-nun [_{NP}[_{CP} Kim-i **ettehakey** yoliha-n] umsik]-ul
 Mimi-TOP Kim-NOM **how** cook-MOD food-ACC
 mekess-ni?
 ate-Q
 ‘What was manner x such that Mimi ate the food that, in x, Kim cooked?’
 ‘What was the result x such that Mimi ate the food that, into x, Kim cooked?’
- (13) a. **eccayse** amwuto kheyikhu-lul mantul-ciahn-ass-ni?
 how anyone cake-ACC make-NEG-PST-Q
 ‘What was the cause x such that, due to x, nobody made any cake?’
- b. *amwuto **ettehakey** kheyikhu-lul mantul-ciahn-ass-ni?
 anyone **how** cake-ACC make-NEG-PST-Q
 *‘What was manner x such that, in x, nobody made any cake?’
 *‘What was the result x such that, into x, nobody made any cake?’

- c. *amwuto kheyikhu-lul **ettehakey** ha-ciahn-ass-ni?
anyone cake-ACC **how** do-NEG-PST-Q
 *‘What was manner x such that, in x, nobody did anything with the cake?’
 *‘What was the result x such that, into x, nobody made the cake into?’
- (14) a. ne-ka **eccayse** pantusi kheyikhu-lul
 you-NOM **how** necessarily cake-ACC
 mantul-cianh-ulswukaiss-ni?
 make-NEG-can-Q
 ‘What is the cause x such that, due to x, necessarily you may not make a cake?’
- b. *ne-ka pantusi **ettehakey** kheyikhu-lul
 you-NOM necessarily **how** cake-ACC
 mantul-cianh-ulswukaiss-ni?
 make-NEG-can-Q
 *‘What is manner x such that, in x, necessarily you may not make a cake?’
 *‘What is the result x such that, into x, necessarily you may not make a cake?’
- c. *ne-ka pantusi kheyikhu-lul **ettehakey**
 you-NOM necessarily cake-ACC **how**
 ha-cianh-ulswukiss-ni?
 do-NEG-can-Q
 *‘What is manner x such that, in x, necessarily you may not do anything with the cake?’
 *‘What is the result x such that, into x, necessarily you may not make a cake?’
- (15) a. **eccayse** motwutul kheyikhu-lul mantul-ess-ni?
how everyone cake-ACC make-PST-CONN
 ‘What was the cause x such that everyone made the cake(s) due to x?’ (how > $\forall$ / * $\forall$ > how)
- b. motwutul **ettehakey** kheyikhu-lul mantul-ess-ni?
 everyone **how** cake-ACC make-PST-CONN

- ‘What was manner x such that everyone made the cake(s) in x?’
 ($\forall > \text{how}$)
 *‘What was the result x such that everyone made the cake(s) into x?’ ($\forall > \text{how}$)
- c. *motwutul kheyikhu-lul ettehakey hay-ss-ni?*
 everyone cake-ACC **how** do-PST-CONN
 *‘What was manner x such that, in x, everyone did something with the cake?’ ($\forall > \text{how}$)
 ‘What was the result x such that, into x, everyone made the cake?’
 ($\forall > \text{how}$)

These data indicate that *eccayse* and *ettehakey* are positioned in two separate domains, the CP-layer and the VP-layer, respectively, similarly to causal and non-causal *ettehkey* (cf. §2.1). Obviously, *eccayse* and *ettehakey* are distinct from each other, since they do not share their phonetic forms.

However, this fact is not trivial. If they are distinct *wh*-expressions in two domains, it opens the possibility that causal and non-causal *ettehkey* are also in fact distinct *wh*-expressions with different syntactic properties, but with undistinguishable phonetic forms. In other words, O. Kim and S. Park’s (2021) assumption (cf. §1) that causal and non-causal *ettehkey* are a single *wh*-adverb positioned in distinct positions with the distinct interpretations and the island sensitivity asymmetry can be reconsidered. If causal and non-causal *ettehkey* are distinct *wh*-expressions, similarly to causal *eccayse* and non-causal *ettehakey*, then the distinct syntactic properties that causal and non-causal *ettehkey* exhibit are unsurprising; they behave in distinct ways, since they are distinct *wh*-expressions. This line of reasoning directly challenges O. Kim and S. Park’s conclusion that the island sensitivity of *ettehkey* is determined solely by the base position, not by the morpho-syntactic make-up.

3. UNSELECTIVE BINDING AND CAUSAL *HOW*

3.1 Introduction to unselective binding analysis

According to Tsai's (1994a, 1999a, 2008) unselective binding analysis, Chinese in-situ *wh*-expressions containing a *wh*-nominal head are not subject to any LF movement; instead, they are unselectively bound by the Q-operator in the CP-layer. Thanks to this binding relation, the in-situ *wh*-nominals can be successfully interpreted at the clausal Left Periphery without covertly moving to the CP-layer. On the other hand, *wh*-adverbs cannot be unselectively bound by the Q-operator, because only nominals qualify as the variables of the Q-operator. For this reason, *wh*-adverbs are subject to LF movement to be properly interpreted, unlike *wh*-nominals or *wh*-adverbials containing a *wh*-nominal (see Ernst 2002:7 for the distinction between *wh*-adverb and *wh*-adverbial). This asymmetry between *wh*-nominals and *wh*-adverbs may seem trivial in matrix environments but it plays a significant role when they are embedded in islands. Consider the following examples:

- (16) a. *_{[NP_{CP} ta **weishenme** xie de] shu] zui youqu?}
 he **why** write MOD book most interesting
 ‘Books that he wrote why are most interesting?’
 b. _{[NP_{CP} **shei** xie de] shu] zui youqu?}
 who write MOD book most interesting
 ‘Books that who wrote are most interesting?’
 c. _{[NP_{CP} ta wei-le **shenme** **yuanyin** xie de] shu] zui}
 he for-PRF **what** **reason** write MOD book most
 youqu?
 interesting
 ‘Books that he wrote for what reason are most interesting?’

As demonstrated in (16), *wh*-nominals are island-insensitive, whereas *wh*-adverbs are island-sensitive in Chinese. This type of asymmetry is not expected under the theoretical assumption that all in-situ *wh*-expressions are uniformly subject to LF movement, since such an assumption will incorrectly predict that every *wh*-expression should be island-sensitive, contrary to the facts (16b, c).

In contrast, the unselective binding analysis clearly accounts for this asymmetry. The Q-operator in the matrix clause unselectively binds the embedded *wh*-nominals (16b, c), and this makes it unnecessary for them

to additionally undergo any LF movement. For this reason, they can be interpreted at the matrix clausal left periphery without violating the Island Constraint. However, this binding relation cannot be established between the Q-operator and any *wh*-adverb, as in (16a), due to its lack of any bindable *wh*-nominal head; a simplex adverb can never contain a nominal head. This forces the *wh*-adverb to undergo an LF movement, which necessarily violates the island constraint. This is why *weishenme* 'why' is island-sensitive.

Tsai's unselective binding analysis has also been applied to other *wh*-in-situ languages, such as Japanese (Fujii et al. 2014) and Vietnamese (C.-Y. Tsai 2009; Phan 2023). As exemplified below, it also successfully applies to Korean:

- (17) a. *_{[NP[CP ku-ka **way** ssu-n] chayk]-i}
 he-NOM **why** write-PST.MOD book-NOM
 ceyil caymiiss-ni?
 most interesting-Q
 'Books that he wrote why are most interesting?'
- b. _{[NP[CP **nwu**-ka ssu-n] chayk]-i ceyil}
 who-NOM write-PST.MOD book-NOM most
 caymiiss-ni?
 Interesting-Q
 'Books that who wrote are most interesting?'
- c. _{[NP[CP ku-ka **ette-n** **iyu-lo** ssu-n]}
 he-NOM **which-MOD** **reason-INST** write-PST.MOD
 chayk]-i ceyil caymiiss-ni
 book-NOM most interesting-Q
 'Books that he wrote for what reason are most interesting?'

3.2 Causal *eccayse* and *ettehkey* as genuine *wh*-adverbs

If this is the case, back to (12a), causal *eccayse* should be regarded as a *wh*-adverb, since it is island-sensitive as is *way* 'why' (17a). Intriguingly enough, we can prove in several ways that causal *eccayse* is in fact a grammaticalized and thus a genuine *wh*-adverb. First of all, *eccayse* is a result of the contraction of *ecci-hay-se* 'how-do-for'. Such a contraction

is one of the most typical features of grammaticalization (cf. Kouteva et al. 2019). Second, the original form *ecci-hay-se* ‘how-do-for’ is ambiguous between causal and manner *how*. This ambiguity arises from the productive verbal suffix *-se* that encodes causal and manner interpretations, as described in the following examples:

- (18) kele-**se** tali-ka aphu-ta
 walk-**for** leg-NOM hurt-DECL
 ‘My leg hurts because of the walk’ (Causal)
- (19) kele-**se** cip-ey ka-n-ta
 walk-**by** home-GOAL go-PRS-DECL
 ‘I will go back home on foot’ (Manner)

In contrary, causal *eccayse* no longer allows for manner interpretation, as mentioned above (10). This indicates that *eccayse* is not merely a phonetic contraction of *ecci-hay-se* ‘how-do-for’, but a grammaticalized *wh*-adverb that does not contain the original verbal suffix *-se*. Specifically, the unavailability of manner interpretation of *eccayse* is due to the fact that it in fact does not contain the verbal suffix *-se*; it is no longer the verbal suffix of a complex phrase, but only a phonetic residue of the original form *ecci-hay-se* ‘how-do-for’. If we regard causal *eccayse* as a grammaticalized *wh*-adverb for these reasons, we can account for its island sensitivity; it is island-sensitive due to its morpho-syntactic make-up simplified into a genuine *wh*-adverb containing no *wh*-nominal head (see also Roberts and Roussou 2003 for the structural simplification of grammaticalization).

We can analyze causal *ettehkey* in the same manner. It is a contraction of *ettehakey* (11), is island-sensitive (3a), and exhibits syntactic properties distinguished from the non-causal *ettehkey* (2b, c) and *ettehakey* (11); it is immune to the intervention effect (6) and negative islands (7), and always scopes over the subject quantifier (8), akin to causal *eccayse* (13-15). Therefore, we regard causal *ettehkey* also as a genuine *wh*-adverb grammaticalized from the non-causal *ettehkey* and *ettehakey*. This of course accounts for its island sensitivity: causal *ettehkey* is island-sensitive

since it is a genuine, grammaticalized *wh*-adverb containing no *wh*-nominal head, as is causal *eccayse*.

3.3 Evidence from Chinese and Japanese

Interestingly enough, Chinese and Japanese causal *how* also exhibit similar patterns. Consider the relevant Chinese examples listed below:

- (20) *zenme* < *zuo* (*shen*)*me*
 how do what
 (cf. Jiang 1994; Lü 1985; Kao 1950)
- (21) **ni* *hui* *chi* [_{NP}[_{CP} *Zhangsan zenme hui zhu*] *de* *cai*]?
 you will eat *Zhangsan how* will cook MOD food
 ‘What will be the cause *x* such that you will eat the food that,
 due to *x*, *Zhangsan* will cook?’
- (22) *Akiu zenme bu xi che*?
 you **how** NEG wash car
 ‘How come *Akiu* won’t wash the car?’ (Tsai 2008:96)
- (23) ***zenme*** *meigeren hui dai yi-ben shu*?
 how everyone will bring one-CL book
 ‘How come everyone will bring one book?’ (Tsai 2008:100)
 (*how* > $\forall$ / * $\forall$ > *how*)

First of all, Chinese causal *zenme* (1) is a contraction of *zuo shenme* ‘do what’, comparable to causal *eccayse* from *ecci-hay-se* ‘how-do-for’ and to causal *ettehkey* from *ettehakey* (11). It is also island-sensitive (21) as are causal *eccayse* (11a) and *ettehkey* (3a). It always scopes over the negation (22) and the subject quantifier (23) similarly to causal *eccayse* (14a, 15a) and *ettehkey* (7a, 8a). Therefore, we can uniformly regard causal *zenme* as a genuine, grammaticalized *wh*-adverb as understood by many previous studies (Tsai 1994a; 1999a; 2008). This not only clearly accounts for its syntactic properties under the unselective binding analysis

but also provides cross-linguistic support for our arguments that causal *eccayse* and *ettehkey* are genuine *wh*-adverbs.

Another piece of substantial evidence comes from Japanese. The Japanese *doosite*, which is interpreted exclusively as causal *how* (Fujii & Takita 2007) exhibits similar properties. Consider the examples below:

- (24) arui-**te** ashi-ga itai
walk-**for** leg-NOM hurt.DECL
'My leg hurts because of the walk' (Causal)
- (25) arui-**te** uchi-e kaeru
walk-**by** home-GOAL go-PRS-DECL
'I will go back home on foot' (Manner)
- (26) *John-wa [NP[CP Mari-ga **doosite** ryoorisi-ta]
John-TOP Mary-NOM **how** cook-PRF.MOD
tabemono]-o tabeta-no?
food-ACC ate-Q
*'What was the cause x such that John ate the food that, due to x, Mary cooked?'
- (27) **doosite** daremo kēki-o tsukura-na-katta-no?
how anyone cake-ACC make-NEG-PST-Q
'What was the cause x such that, due to x, nobody made any cake?'
- (28) omae-wa **doosite** kinoo kēki-o tsukura-na-katta-no?
you-TOP **how** yesterdaycake-ACC make-NEG-PST-Q
'What was the cause x such that, due to x, you did not make any cake yesterday?'
- (29) **doosite** minna kēki-o tsukura-na-katta-no?
how everyone cake-ACC make-NEG-PST-Q
'What was the cause x such that everyone made the cake(s) due to x?' (how > ∀ / *∀ > how)

Similarly to the Korean causal *eccayse* from *ecci-hay-se* ‘how-do-for’, *doosite* originated from *doo-si-te* ‘how-do-for’. Also, akin to the Korean suffix *-se* (12-13), the Japanese verbal suffix *-te* is ambiguous between causal and manner interpretations (24-25), whereas *doosite* no longer permits manner interpretation, similarly to Korean *eccayse* (10b, c). It is island-sensitive (26) and scopes over SBEs (27), negations (28), and subject quantifiers (29), similarly to causal *eccayse* (13a, 14a, 15a) and *ettehkey* (6a, 7a, 8a). Therefore, we should regard causal *doosite* also as a genuine, grammaticalized *wh*-adverb as suggested by Fujii and Takita (2007).

Therefore, both Japanese causal *doosite* and Chinese causal *zenme* are grammaticalized genuine *wh*-adverbs. These Chinese and Japanese data cross-linguistically support our current argument that the Korean causal *eccayse* and *ettehkey* are grammaticalized genuine *wh*-adverbs.

3.4 Position of causal *how*

Now, we know that Korean causal *eccayse* and *ettehkey*, Chinese causal *zenme*, and Japanese causal *doosite* are uniformly grammaticalized genuine *wh*-adverbs subject to the island constraints due to their simplex non-nominal morpho-syntactic make-up. The scope relations they establish with SBEs, negations, and subject quantifiers further indicate that they are CP-layer adverbs (cf. §2.1 for Korean and §3.3 for Chinese and Japanese). However, under the split-CP of the cartographic approach (Rizzi 1997; 2004), this level of description of their position may not be sufficient. In this paper, we suggest that these adverbs are positioned at Spec,Cau(se)P accommodated in the clausal Left Periphery, based on the distributional and semantic evidence of causal *how* in Korean and Chinese.

First of all, in Chinese, causal *zenme* must precede the verbal complex containing the outer light verb CAUSE, which precedes the agent subject, as illustrated in (30) (Tsai 2014; 2015). When CAUSE is absent, *zenme* preceding the verbal complex is ambiguous between causal and manner *how* (31) due to the absence of any clear delimiter.

- (30) na-ba dao {zenme} qie-de-Ø *{zenme} ni
 that-CL knife **how** cut-RES-CAUSE **how** you

zhi maohan ne?
 continuously sweat Q
 ‘How come that knife made you cut such that you kept
 sweating?’ (Tsai 2015:22)

- (31) Akiu **zenme** qie-Ø na-ba dao ne?
 Akiu **how** cut-USE that-CL knife Q
 a. ‘How come Akiu cut with that knife?’
 b. ‘How did Akiu cut with that knife?’ (Tsai 2015:22)

Also, causal *zenme* encodes eventuality causation between a cause, which is being asked with *zenme*, and an effect, which is, for example, “Akiu’s cutting with that knife” (31a). This indicates that causal *zenme* semantically scopes over the whole event including the agent subject *Akiu* ‘Akiu’ although the subject may sometimes undergo topicalization to a higher position as in (31). This scope relation is further supported by the interaction between causal *zenme* and subject quantifiers (23). Furthermore, causal *zenme* exhibits a strong tendency to precede IP-domain adverbs, but follow the SAP-domain adverbs as exemplified below:

- (32) a. ni {zenme} yijing *{zenme} zuo-le ge dangao
 you how already how make-PRF CL cake
 ‘What was the cause x such that, due to x, you already made a
 cake?’
 b. ni {zenme} changchang *{zenme} zai zuo dangao
 you how always how PROG make cake
 ‘What is the cause x such that, due to x, you are always making
 cakes?’
 c. ni {zenme} yiding *{zenme} hui zuo dangao
 you how necessarily how will make cake
 ‘What will be the cause x such that, due to x, you will necessarily
 make a cake?’
- (33) *{zenme} laoshishuo {zenme} ni jingchang yao zuo
 how frankly how you always must make

dangao
cake

‘Frankly, what can be the cause x such that, due to x, you always have to make cakes?’

From the perspective of the semantic interpretation, causal *zenme* is distinguished from reason *weishenme* ‘why’ as exemplified below (cf. Tsai 2008; 2019):

- (34) a. tiankong **zenme** shi lande? [causal]
sky **how** be blue
‘How come the sky is blue (as the typhoon is approaching)?’
b. tiankong **weishenme** shi lande? [reason]
sky **why** be blue
‘Why is the sky blue (a scientific account is in order)?’

(34a) is a question about an individual event, which is counter-intuitive in this case. On the other hand, (34b) is a question about the scientific reason for the general fact that the sky is blue. For this reason, (34b) is a question that a science teacher may ask to his/her students in the classroom, whereas (34a) is not. Based on these syntactic and semantic properties of causal *zenme*, we can conclude that causal *zenme* is situated at the specifier position of CauP, the outer causative light verb phrase above FinP (cf. Tsai 2024).

Unfortunately, Korean is a strictly head-final language, making it difficult to use such a verbal complex as a direct delimiter of causal *how*. Nonetheless, the Korean causal *how* exhibits strict distributions in regard to other types of adverbs as demonstrated below:

- (35) a. ne-nun {ettehkey} imi *{ettehkey} kheyikhu-lul³
you-TOP how already how cake-ACC

³ A reviewer kindly informed us the fact that causal *ettehkey* may follow, for example, some IP and VP-domain constituents as exemplified below:

- (i) ne-nun ecey ku kheyikhu-lul ettehkey mantul-ess-ni?
you-TOP yesterday the cake-ACC how make-PST-Q
‘What was the cause x such that, due to x, you made the cake yesterday?’

- mantul-ess-ni?
make-PST-Q
‘What was the cause x such that, due to x, you already made a cake?’
- b. ne-nun {ettehkey} enceyna {*ettehkey} kheyikhu-lul
you-TOP how always how cake-ACC
mantul-koiss-ni?
make-PROG-Q
‘What is the cause x such that, due to x, you are always making cakes?’
- c. ne-ka {ettehkey} pantusi {*ettehkey} kheyikhu-lul
you-NOM how necessarily how cake-ACC
mantul-ni?
make-Q
‘What will be the cause x such that, due to x, you will necessarily make a cake?’
- (36) {*ettehkey} solcikhi {ettehkey} ne-ka enceyna
how frankly how you-NOM always

We attribute the grammaticality of such surface word order to the scrambling which is highly frequent and free in Korean. This argument is supported by the following two examples:

- (ii) * ne-nun imi ku kheyikhu-lul ettehkey mantul-ess-ni?
you-TOP already the cake-ACC how make-PST-Q
‘What was the cause x such that, due to x, you already made the cake?’
- (iii) ne-nun ku kheyikhu-lul ettehkey imi mantul-ess-ni?
you-TOP the cake-ACC how already make-PST-Q
‘What was the cause x such that, due to x, you already made the cake?’

Note that it is extremely difficult to interpret *ettehkey* in (ii) as causal *how*; instead, it can only be interpreted as either of the two non-causal *how*. This is due to the relative positions between the IP-domain adverb *imi* ‘already’ and causal *ettehkey*, which are ungrammatical as demonstrated in (35a). This is the reason why, regardless of the scrambling of the internal argument, (ii) is ungrammatical. On the other hand, (iii) is grammatical despite the scrambled internal argument over the CP and IP-domain adverbs, i.e. *ettehkey* and *imi*. This result matches with Ko’s (2008) argument that the Korean scrambling may aim for the CP-domain. Therefore, we also regard (i) as an example of scrambling, following O. Kim and S. Park’s (2021) argument that causal *ettehkey* is a CP-domain adverb.

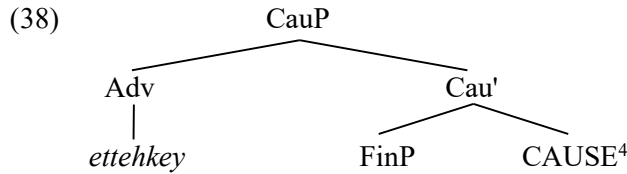
kheyikhu-lul mantul-ni?
 cake-ACC make-Q
 ‘Frankly, what can be the cause x such that, due to x, it is
 always you that makes cakes?’

As demonstrated in (35-36), causal *ettehkey* strictly precedes the IP-domain adverbs, but follow the SAP-domain adverbs (cf. Cinque 1999). This relative position coincides with O. Kim and S. Park’s (2021) conclusion that causal *ettehkey* is a CP-domain adverb (cf. §2.1). Furthermore, as discussed above (cf. 2.2), causal *ettehkey* also scopes over SBEs (6a), negations (9a), and subject quantifiers (10a), similarly to another CP-domain adverb *way* ‘why’ (Ko 2005; 2006).

Semantically, causal *ettehkey* behaves similarly to Chinese causal *zenme* discussed above (34). Consider the following examples:

- (37) a. hanul-i ettehkey phalah-ni? [causal]
 sky-NOM how blue-Q
 ‘How come the sky is blue (as the typhoon is approaching)?’
 b. hanul-i way phalah-ni? [reason]
 sky-NOM why blue-Q
 ‘Why is the sky blue (a scientific account is in order)?’

The reason question, i.e. (37b), is more likely to be asked by a science teacher in the classroom, whereas (37a) asks a question about an individual event that is caused by another event, e.g. a typhoon. We capture these syntactic and semantic properties with CauP in the cartographic manner (Rizzi 1997; Cinque 1999), similarly to the Chinese cases (Tsai 2014; 2015; 2019), as demonstrated below:



⁴ One of the reviewers kindly suggested the possibility of *toy* ‘become’ being the Cause head. We also have seriously considered this possibility as one of our determining

4. NON-CAUSAL *HOW* AS NON-ADVERBS

In Section 2, we confirmed that causal and non-causal *ettehkey* are possibly distinct *wh*-expressions. In Section 3, we accounted for the various syntactic properties of causal *ettehkey*, including its island sensitivity, by regarding it as a genuine, grammaticalized *wh*-adverb under the unselective binding analysis. In this section, we discuss the syntactic properties of non-causal *ettehkey*, such as island insensitivity, base positions, exponents, and so on, uniformly under the unselective binding analysis but also from the perspective of Distributed Morphology.

4.1 Two distinct positions of the two non-causal *how*

supporting arguments for the Cause head. However, we eventually decided to reserve this possibility for future studies for the following reasons. First of all, causal *ettehkey* is distinguished from the reason *way* ‘why’ as demonstrated in (37), but the so-called *become*-construction with *toy* is compatible with both of them (i-ii) and does not even have to be associated with any *wh*-expression (iii) as exemplified below:

- (i) ne-nun way kheyikhu-lul mantul-key toy-ss-ni?
you-TOP why cake-ACC make-RES become-PST-Q
‘Why did you end up making a cake?’
- (ii) ne-nun ettehkey kheyikhu-lul mantul-key toy-ss-ni?
you-TOP how cake-ACC make-RES become-PST-Q
‘How did you end up making a cake?’
- (iii) ne-nun kheyikhu-lul mantul-key toy-ss-ni?
you-TOP cake-ACC make-RES become-PST-Q
‘Did you end up making a cake?’

Furthermore, as (37b) demonstrates, causal *ettehkey* does not require the *become*-construction either. Also, (ii) only indicates that the subject resulted (as resultative marker *-key* indicates) in having a plan to make a cake; it does not indicate that the making event has already taken place. On the contrary, the following example with causal *ettehkey*, but without *toy* ‘become’ indicates that the making event has happened:

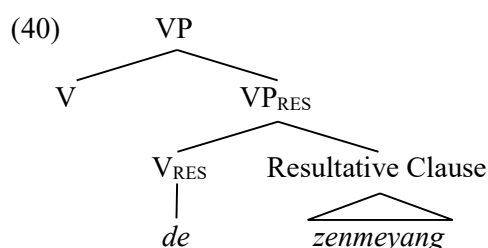
- (iv) ne-nun ettehkey kheyikhu-lul mantul-ess-ni?
you-TOP how cake-ACC make-PST-Q
‘How come you made a cake?’

This indicates that causal *ettehkey* and the *become*-construction are not in the necessary-sufficient relation. In other words, we cannot regard *toy* ‘become’ as the dedicated functional head for causal *ettehkey*. For this reason, unfortunately, we do not regard *toy* ‘become’ as the Cause head in this paper.

(2c) indicates that resultative *ettehkey*, one of the two non-causal *ettehkey*, is embedded in the deepest position of the VP-layer. We describe this tendency even more clearly in morpho-syntactic and cross-linguistic terms. First of all, in Chinese, frequency adverbs regularly suppress resultative *zenmeyang*. Consider the following examples:

- (39) a. Zhangsan **zenme(yang)** zuo liangci?
 Zhangsan **how** make twice
 ‘What is manner x such that Zhangsan makes something twice in x?’
 *‘What is the result x such that Zhangsan makes something into x, twice?’
- b. *Zhangsan hui ba dangao zuo de
 Zhangsan will BA cake make DE
 {**zenmeyang**} liangci {**zenmeyang**}
 twice **how**
 *‘What will be manner x such that Zhangsan will make something twice in x?’
 *‘What will be the result x such that Zhangsan will make something into x, twice?’

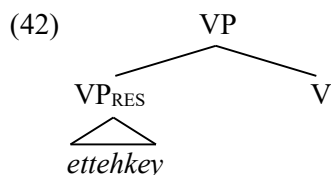
The reason why resultative *how* is suppressed by frequency adverbs in Chinese (39) is that resultative *how* and frequency adverbs are both so-called “inner adjuncts” (Huang 1984) and hence compete for the identical complement position of the lowest verb head as described in the following structure (cf. Tsai 2024; see also Ramchand 2008 and Ramchand and Svenonius 2014 for V_{RES}):



Interestingly, Korean resultative *ettehkey* and frequency adverbs also exhibit mutual incompatibility as described below:

- (41) a. ne-nun ecey ku kheyikhu-lul **ettehkey** twupen
 you-TOP yesterday the cake-ACC **how** twice
 mantul-ess-ni?
 make-PST-Q
 ‘What was manner x such that, in x, you made the cake twice
 yesterday?’
 *‘What was the result x such that, into x, you made the cake
 twice yesterday?’
- b. *ne-nun ecey ku kheyikhu-lul twupen **ettehkey**
 you-TOP yesterday the cake-ACC twice **how**
 mantul-ess-ni?
 make-PST-Q
 *‘What was manner x such that, in x, you made the cake twice
 yesterday?’
 *‘What was the result x such that, into x, you made the cake
 twice yesterday?’

As (41a) indicates, the frequency adverb *twupen* ‘twice’ suppresses resultative interpretation of *ettehkey* that is available in absence of the frequency adverb (2c). The unavailability of resultative *ettehkey* in presence of a frequency adverb can be clearly accounted for by arguing that resultative *ettehkey* and the frequency adverb *twupen* ‘twice’ compete for the identical complement position of the lowest verb head as described in (42), akin to the Chinese structure (40). This accounts not only for the incompatibility of the two, but also for the low distribution of resultative *ettehkey* (2c).



Also, *ettekha(y)* ‘do how’, another Korean *how*-expression, is a contraction of resultative *ettehkey* and the light verb *ha(y)* ‘do’. This form is typically used to ask about resultative state of the internal argument as described in the following example:

- (43) ne ku kheyikhu ettekhay-ss-ni?
 you the cake do.how-PST-Q
 ‘What did you do to the cake (how is the cake now)?’

Such a contraction typically indicates the adjacency of the two relevant constituents, i.e., resultative *ettehkey* and the light verb *ha(y)* ‘do’, supporting (42) (see also Nam 1988 and Nam and Cho 1993 for further examples on the constituency of resultatives).

On the other hand, manner *zenme* (39a) and *ettehkey* (41a) are not influenced by the presence of frequency adverbs. In semantics, they are not additional predicates about the internal argument either, in contrast with resultative *how*. Instead, they are predicates of the underlying event argument (Tsai 2008). In addition, they are both incompatible with sentient verbs whose subject cannot be an agent in both Chinese (44-45) and Korean (46-47) (Chinese examples from Tsai 2008:92):

- (44) Akiu **zenme** xihuan hua, jiu guli dajia zhong
 Akiu **how** like flower then encourage people plant
 a. ‘How come Akiu likes flowers, and then encouraged people to plant them?’
 b. #‘In what manner does Akiu likes flowers, and then encouraged people to plant them?’ (Transitive sentient verb)
- (45) Akiu **zenme** ku-le
 Akiu **how** cry-INC
 a. ‘How come Akiu started to cry?’
 b. #‘In what manner did Akiu start to cry?’ (Unergative sentient verb)
- (46) John-i **ettehkey** kkoch-ul cohaha-ni?
 John-NOM **how** flower-ACC like-Q

- a. ‘How come John likes flowers?’
 b. #‘In what manner does John likes flowers?’ (Transitive sentient verb)

- (47) John-i **ettehkey** wul-ess-ni?
 John-NOM **how** cry-PST-Q
 a. ‘How come John started to cry?’
 b. #‘In what manner did John cry?’ (Unergative sentient verb)

This means that manner *how* in both languages are requires an agent, which is introduced by the *vP*. This, together with the other syntactic properties, implies that manner *how* are adjuncts in the peripheral area of the *vP*, (see also Tsai 2008; Fujii et al. 2014 for further relevant discussions on Chinese and Japanese manner *how*), as illustrated below:

- (48)
-
- ```

graph TD
 vP1[vP] --- ettehkey[ettehkey]
 vP1 --- vP2[vP]

```

#### 4.2 The structures of the two non-causal *how*

As for manner *ettehkey*, the form *ettehkey* has traditionally been analyzed to consist of *etteh* and *-key*. The suffix *-key* is an adverbializer for adjectives. For example, *ppalu* ‘quick’ can be adverbialized by *-key* into a manner adverb *ppalu-key* ‘quickly’. (cf. Martin 1992). Therefore, manner *ettehkey* (48) can be analyzed as follows:

- (49)
- 
- ```

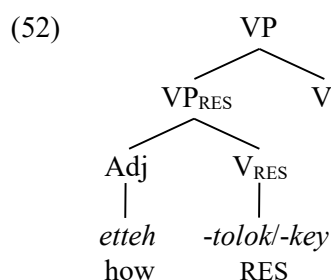
graph TD
  vP1[vP] --- AdvP[AdvP]
  vP1 --- vP2[vP]
  AdvP --- Adj[Adj]
  AdvP --- Adv[Adv]
  Adj --- etteh[etteh]
  etteh --- how[how]
  Adv --- key[-key]
  key --- ADV[ADV]
  
```

We can analyze resultative *ettehkey* in a similar manner, but in this case, we understand *-key* as the head of the VP_{RES} (42) that is comparable to Chinese *de* (40). We make this argument based on another exclusively resultative *how*, *etteh-tolok*, which must follow the internal argument (50) akin to resultative *ettehkey*. The suffix *-tolok* productively forms resultative predicates and can be attached not only to adjectival predicates such as *etteh* ‘how’, but also to verbal predicates (cf. Martin 1992). Similarly to resultative *ettehkey* (41), *etteh-tolok* is also blocked by frequency adverbs (51).

- (50) a. *ne-nun ecey **etteh-tolok** ku kheyikhu-lul
 you-TOP yesterday **how** the cake-ACC
 mantul-ess-ni?
 make-PST-Q
 *‘What was manner x such that, in x, you made the cake
 yesterday?’
 *‘What was the result x such that, into x, you made the cake
 yesterday?’
- b. ne-nun ecey ku kheyikhu-lul **etteh-tolok**
 you-TOP yesterday the cake-ACC **how**
 mantul-ess-ni?
 make-PST-Q
 *‘What was manner x such that, in x, you made the cake
 yesterday?’
 ‘What was the result x such that, into x, you made the cake
 yesterday?’
- (51) *ne-nun ecey ku kheyikhu-lul twupen **etteh-tolok**
 you-TOP yesterday the cake-ACC twice **how**
 mantul-ess-ni?
 make-PST-Q
 *‘What was the result x such that, into x, you made the cake
 twice yesterday?’

The incompatibility of *etteh-tolok* and frequency adverbs (51) indicates that *etteh-tolok* is also the complement of the lowest verb head, i.e. the

VP_{RES} in (42), akin to resultative *etteh-key*. Since resultative interpretation is certainly encoded by *-tolok*, it can be argued that *-tolok*, together with resultative *-key*, heads the VP_{RES}, akin to Chinese *de* (40), as described below:



Now, we can also account for the ambiguity of *-key* between manner and resultative interpretations; it encodes either the Adv head (49) or the V_{RES} head (52), which are introduced in two different positions, thereby triggering the relevant ambiguity.

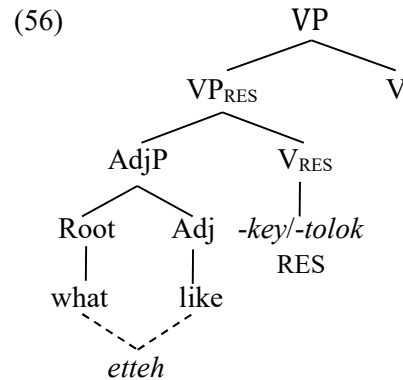
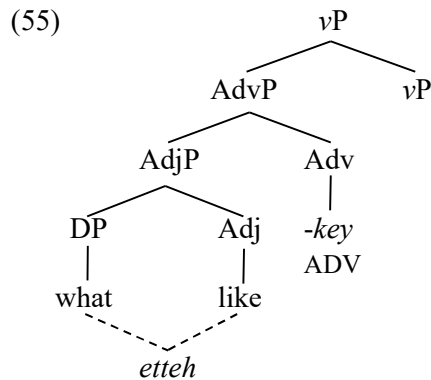
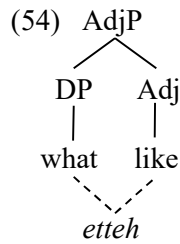
4.3 Island insensitivity of non-causal *how*

Now, we are ready to account for the island insensitivity of the two non-causal *ettehkey*. In the last subsection, we treated *etteh* as an adjective for *how* (49, 52). Interestingly, *etteh* can further be analyzed into two constituents from the perspective of its etymology. The form *etteh* dates back to the Middle Korean non-causal *how*, *:es-téhò* ‘what-like’ (53d) (cf. Nam 1997). This Middle Korean form pairs with other kinds of demonstrative non-*wh*-expressions as in (53) (see Martin 1996 for the reconstruction of /t/ out of intervocalic /l/):

- | | |
|--|---|
| (53) a. í-téhò
this-like
like this | b. kù-téhò
that-like
like that (medial) |
| c. tyé-téhò
that-like
like that (distal) | d. :es-téhò
what-like
like what |

(53a-c) mean ‘like this’, ‘like that (medial)’, and ‘like that (distal)’, indicating that the suffix *-téhò* should be a similative adjective ‘like’. The *wh*-form in question, *:es-téhò*, also contains this root and it is construed as ‘like what’. This clearly signals that *:es* must be regarded as *what*, an indeterminate noun in Kuroda’s (1965) terminology (see also Jang 2013 and Lee 2009 for the etymology of Middle Korean *mùès* ‘what’ from *mùsú kès* ‘which thing’, comparable to the formation of *:es* ‘what’ from *ènú kès* ‘which thing’). In other words, *:es-téhò* must be analyzed as an AdjP headed by *-téhò* ‘like’ with its complement *:es* ‘what’, not as an adjective.

We can analyze the modern form *etteh* ‘how’ in the same manner, as an AdjP headed by a similative adjective *-teh* ‘like’. However, the modern form *et* ‘what’ appears exclusively in this specific context in which it is the complement of the similative adjective *-teh* ‘like’. To account for this fact, we understand the modern form *etteh* as one exponent for two formatives, i.e., *what* and *like*, as described in (54). Based on (54), (49) and (52) can be respectively reanalyzed as (55) and (56).



4.4 Comparisons with other languages

Japanese manner *doo(-yatte)* ‘how’ is also island-insensitive, similarly to Korean non-causal *ettehkey* (cf. Nishigauchi 1990; Ura 1993). Consider the Japanese examples below, comparable to the Korean examples (3a):

- Under the unselective binding analysis, Fujii and Takita (2007) account for the island insensitivity of *doo*-(*yatte*) ‘how’ by assuming an empty nominal head for *doo* (58). *Doo* is contained within a DP since *doo* follows the so-called *ko-so-a-do* ‘this-that-that-which’ (*wh*-)determinative pattern (59), similarly to Korean *ettehkey* (53).

- (59) a. koo
like.this.ADV
like this
c. aa
like.that.ADV
like that (distal)
- b. soo
like.that.ADV
like that (medial)
d. doo
like.what.ADV
like what

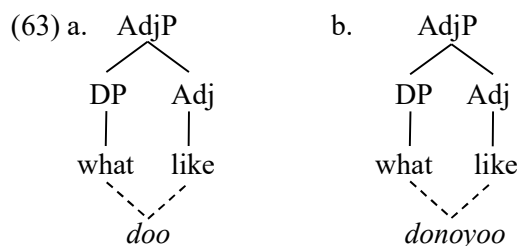
However, the structure (58) is based on some stipulations. First of all, *doo* is traditionally understood as an adverbial, which is not usually expected in a DP domain. Second, the presence of the empty nominal head is still not supported by any independent evidence other than the theoretical inference based on the unselective binding analysis.

We suggest that *doo* ‘how’ is etymologically and morphologically analyzed to have a structure similar to (54). Consider the adjective *doo-da* ‘be how’ (60a). It follows the so-called *na*-adjective conjugation. In other words, the predication suffix *-da* must be substituted by the modificational suffix *-na* (60b) in order to modify nominals. However, unlike regular *na*-adjectives, e.g. (61b), *doo* in the adjectival form (60a) must undergo an irregular readjustment into *don* (60b). Interestingly, Japanese has another *how*-expression (62) synonymous to (60), which, contrary to *doo*, follows the *na*-adjective conjugation rule.

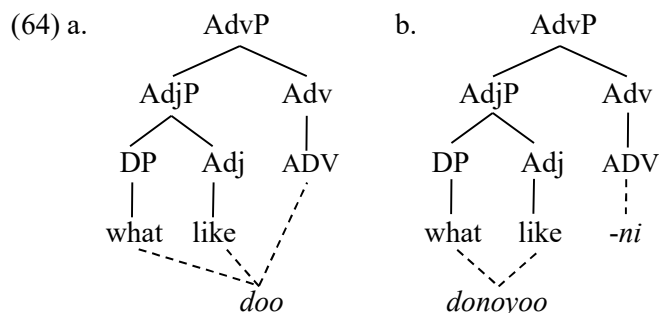
- | | | |
|---|--|--|
| (60) a. doo -da
how-PRED
to be how | b. don -na (*doo-na)
how-MOD
which, what kind of | c. doo (*-ni)
how-ADV
how |
| (61) a. anzen-da
safe- PRED
to be safe | b. anzen-na
safe-MOD
safe | c. anzen-ni
safe-ADV
safely |
| (62) a. donoyoo -da
which.like- PRED
to be how | b. donoyoo -na
which.like-MOD
which, what kind of | c. donoyoo -ni
which.like-ADV
how |

This *wh*-expression (62a) does not exhibit any readjustment of *donoyoo* when it takes the modificational form (62b). In addition, *donoyoo* in (62a) and (62b) contains all the phonemes of *doo* (60a) and *don* (60b), which implies that the irregular readjustment of *doo* into *don* is presumably due to contractions of (62a) into (60a) and (62b) into (60b), respectively. From the perspective Distributed Morphology, such contractions indicates that *doo* and *don* are two distinct exponents of the same formatives, i.e. *donoyoo*. Considering the fact that *noyoo* ‘like’ exclusively c-selects for nominals, such as *dore* ‘what’ or ‘which thing’, we can further analyze

donoyoo also as an exponent of two formatives, i.e., *what* and *like*, akin to (54), as described below:



Interestingly, the adverbials, (60c) and (62c), also exhibit similar patterns. The adverbial *doo* cannot co-occur with the regular adverbial marker *-ni* (61c), indicating that the adverbial *doo* (60c) is an exponent of three formatives in this case as illustrated in (64a), which is comparable to the structure (64b) for (62c) with the overt Adv head *-ni*.



If (64a) correctly represents the internal structure of *doo*, then its island insensitivity of *doo(-yatte)* ‘how’ (57) is due to the presence of the *wh*-nominal head *what*, which is spelled out by *doo* along with the Adj and Adv heads (64a). In this way, we can develop Fujii and Takita’s (2007) nominal analysis (58) without assuming any empty category. More importantly, the fact that we can account for not only Korean island-insensitive non-causal *how*, but also Japanese island-insensitive non-causal *how* in a similar manner, strongly supports our *wh*-nominal analysis of Korean non-causal *how* (55-56).

4.4.2 Chinese non-causal *how*

The Chinese instrumental non-causal *how*, *zenmeyang*, has also been analyzed in a similar way (Stepanov and Tsai 2008). *Zenmeyang* ‘how’ is ambiguous between the instrumental and manner interpretations (65). Interestingly, the instrumental *zenmeyang* is island-insensitive (66a), whereas manner *zenmeyang* is not (66b).

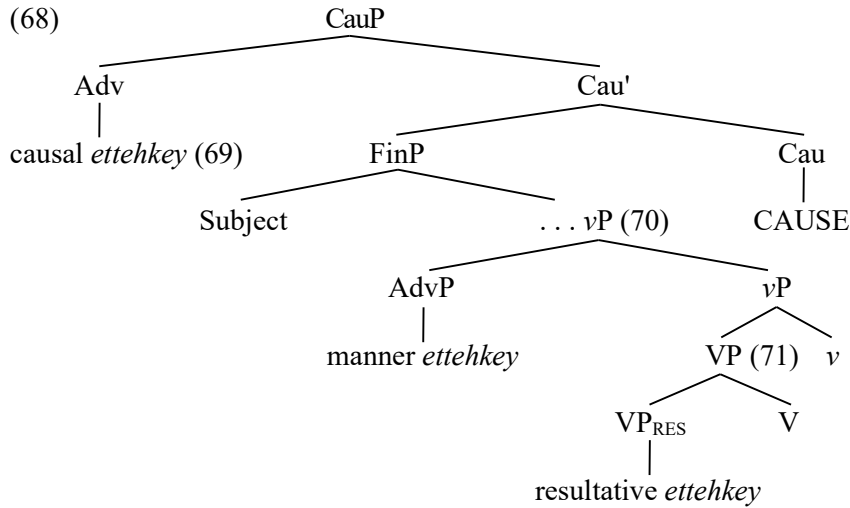
- (65) ta shang-ci **zenmeyang** dun niurou?
 she last-time **how** stew beef
 a. ‘By what means did she stew beef last time?’
 b. ‘In what manner did she stew beef last time?’ (Tsai 1994b:83)
- (66) ni zui xihuan [NP[CP ta **zenmeyang** dun]] de
 you most like she **how** stew DE
 niurou]?
 beef
 a. ‘What is the means x such that you like best [beef [which she stewed by x]]?’
 b.#‘What is manner x such that you like best [beef [which she stewed in x]]?’ (Stepanov and Tsai 2008:596)
- (67) a. zhe-yang b. na-yang c. zenme-yang
 this-manner that-manner how-manner
 like this like that how

Stepanov and Tsai (2008) attribute the island-insensitivity of the instrumental *zenmeyang* (66a) to the presence of a nominal head *yang* ‘manner’, which forms a *wh*-nominal together with *zen* ‘how’. *Zenmeyang* is also paired with other determinative non-*wh*-expressions (67), similarly to the Korean (53) and Japanese (59) examples. On the other hand, Stepanov and Tsai regard the island-sensitive manner *zenmeyang* as a grammaticalized *wh*-adverb from the perspective of the unselective binding analysis in the way in which we analyzed the Korean, Chinese and Japanese causal *how* as genuine *wh*-adverbs. They support their idea by comparing the “bleached” construal of manner *how* with the “unbleached”

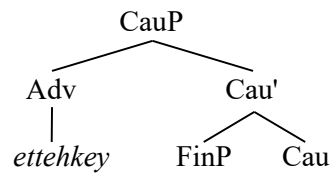
construal instrumental *how* (Stepanov and Tsai 2008:fn.4). Although the Chinese manner *zenme(yang)* is distinct from the Korean manner *ettehkey* in this sense, the fact that island-insensitivity of an in-situ *wh*-adverbial is associated with the presence of a *wh*-nominal head also in Chinese strongly supports our argument regarding the Korean non-causal *ettehkey*.

5. CONCLUSION

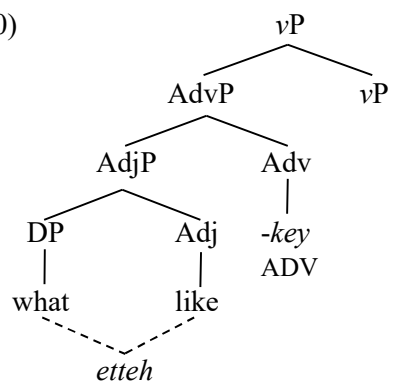
Accordingly, the three types of *ettehkey* exhibit the distributions along the whole clause (68), which is comparable to the distributions of Chinese *how* (1). Causal *ettehkey* is a grammaticalized, genuine, and hence unselectively unbindable *wh*-adverb base-generated at Spec,CauP (cf. §3). On the other hand, manner and resultative *ettehkey* are respectively an AdvP adjoined to vP (70) and a VP_{RES}, the complement of the lowest verb head (71). Both of them contain unselectively bindable *wh*-nominal heads, and this licenses their island-insensitivity (cf. §4). In this way, we successfully deny O. Kim and S. Park's (2021) argument that the island sensitivity of *ettehkey* is solely determined by its position, not by its morpho-syntactic make-up (cf. §2).



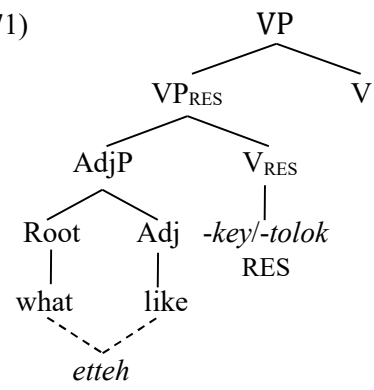
(69)



(70)



(71)



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關於韓語「如何」的跨語言研究

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韓語的 *ettehkey* 「怎麼」依其“詮釋高度” (Hacquard 2006) 被詮釋為因果、方式或結果的「怎麼」。當它被詮釋為方式或結果的「怎麼」時，它對強孤島效應免疫，而當被詮釋為因果的「怎麼」時，則不然。這一觀察對關於原位疑問詞的名副詞對立泛化 (Tsai 1999a; Fujii 等 2014) 提出挑戰，因為該詞似乎不包含名詞性成分。不過本文認為無擇約束分析可以透過將韓語的 *ettehkey* 分析成一組功能詞的方式擴展到對韓語「怎麼」的分析。我們的結論如下：韓語的因果「怎麼」已演變成位於左緣結構的真正的副詞，而方式「怎麼」和結果「怎麼」在動詞組領域內仍然保持有包含不定名詞 (Kuroda 1964) 的複雜結構。我們還借助來自漢語和日語的跨語言證據，進一步支持這一結合製圖理論和分布式型態學的新研究方法。

關鍵字：比較句法、無擇約束、句法化、句法製圖理論、分布式型態學、原位疑問詞